ADAMAS: an open, tested framework for diamond electronics and room-temperature nitrogen-vacancy quantum processors

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github.com/Normansrule/adamas-diamond-framework

Preprint, generated from the repository

Abstract

Diamond combines the largest breakdown field and thermal conductivity of any practical semiconductor with a defect, the nitrogen-vacancy (NV) center, whose electron spin is a qubit at room temperature. ADAMAS is an open framework that follows one path from wafer to quantum processor with executable, cited models: 556 references (500 verified against Crossref), 53 regenerable figures, a process design kit with a placed 4-bit processor, circuit-level quantum error correction, and hands-on experiments from US$100 upward. Three results stand out. (i) Measured hydrogen-terminated diamond transistors already beat silicon's one-dimensional limit by ∼89×, and in an 800 V traction inverter an ideal diamond switch loses 3.8 W against 29.7 W for silicon carbide. (ii) Evaluated with the published parameters of the 2013 room-temperature entangled pair, coherence alone would permit a Bell fidelity of 0.998; the measured 0.67 corresponds to a charge-state preparation probability of 0.75, identifying preparation, not coherence, as the bottleneck. (iii) Circuit-level simulations with idle errors during millisecond readout show that surface-code scaling survives only while readout stays below about a tenth of the nuclear-memory coherence time, that a bias-tailored XZZX code extends this about tenfold, and that a factoring-scale machine fits on a 5.7 mm die but runs for years. Every number in this paper is generated from the package.

1 Introduction

Silicon's dominance rests on process maturity, not material merit. For power conversion, high-temperature electronics, and room-temperature quantum technology, diamond's figures of merit are extreme: its Baliga figure of merit for power switches is 48,976× silicon's and its thermal conductivity 14.7× [1, 2]. Its obstacles are equally concrete: boron acceptors sit 0.37 eV deep, so under 1% ionize at room temperature [3]; single-crystal wafers only recently reached 3 inches [4]; and logic has been p-channel only until the first n-channel MOSFET in 2024 [5].

ADAMAS treats these facts as inputs to models that can be run, tested, and falsified. The contribution is not a new device but a connected, reproducible chain: materials → process → circuits → qubits → error correction → machine size → experiments, with each link checked against published data or an independent tool (ngspice, Icarus Verilog, Yosys, KLayout, Stim, PyMatching).

2 Power devices and converters

The unipolar limit $R_{on,sp} = 4BV^2/(\varepsilon\mu E_c^3)$ [1] places ideal diamond far below every other material (Fig. 1). Measured lateral hole-gas transistors reach 3659 V [8] and, at 2568 V, 7.54 mΩ cm² [7], about 89× better than silicon's limit. For an area-optimized hard-switched leg, conduction and switching losses equalize and the loss is $P = 2IV\sqrt{fR_{on,sp}C_{oss,sp}/2}$ [9]. Table 1 shows the result for an 800 V, 300 A, 20 kHz traction inverter; bulk-doped diamond additionally improves with temperature as boron ionizes [10].

Switch (1.2 kV class)	Loss per switch (W)
Silicon, ideal	209.5
Silicon carbide, ideal	29.7
Gallium nitride, ideal	20.6
Diamond, ideal	3.8
Diamond, 2022 device scaled	89

Table 1: Area-optimized switch loss in an 800 V, 300 A, 20 kHz inverter leg (first order: no gate charge or reverse recovery).

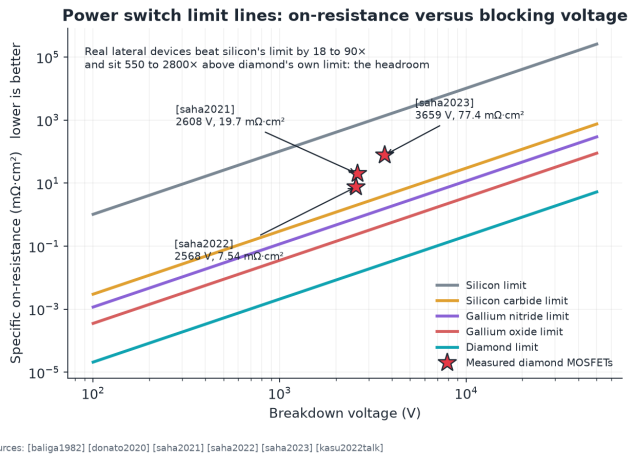


Figure 1: One-dimensional on-resistance limits and measured hydrogen-terminated diamond MOSFETs [6, 7, 8]. Real devices sit above diamond's own limit but below silicon's.

3 Process, design kit, and a placed processor

The hydrogen-terminated flow (Fig. 2) creates the channel by surface transfer doping [11], protects it with gold immediately, isolates devices with one oxygen-plasma step, and stabilizes the hole gas with atomic-layer-deposited Al_2O_3 at $\leq$150 °C (with NO_2) or ∼450 °C (for 400 °C operation) [12, 13, 14]. Its thermal budget forces growth and the NV anneal (up to 1100 °C [15]) ahead of all metal. The traveler gives parameters with sources, pass criteria, safety, and failure modes for each step, and an interactive run log. On this process, the design kit PDK-0 supplies layers, rules, a GDS monitor die, standard cells, LEF abstracts, and a timed Liberty file; the DIA-4 processor synthesizes to 323 cells and places in about 1 mm^2 with pads. An emulator matches its Verilog cycle for cycle on random programs.

4 The two-qubit bottleneck

With the published pair's parameters, 25 nm spacing, 4.93 kHz single-quantum coupling (a 25 μs double-quantum gate), and echo times of 150 and 514 μs [16], the coherence-limited Bell fidelity is 0.998. The measured 0.67 corresponds to a preparation probability $q = 0.75$ and the later 0.82 [17] to $q = 0.87$ (Fig. 3). Both match the NV^- fraction under green light [18]. A five-level rate model with measured rates [19] gives a readout contrast of 45% in a 300 ns window and 82% spin polarization; charge-state verification before the gate is therefore the highest-leverage experimental change.

5 Error correction with slow readout

NV readout takes about a millisecond [20, 21], during which data qubits held in nuclear spins idle. Circuit-level simulations with Stim and PyMatching [22, 23] (0.3% link, 1% readout, 0.2% preparation errors) validate a threshold near 1.2% for uniform noise and give an error-suppression factor $\Lambda = 4.03$ at 0.1 ms readout, 3.56 at 1 ms, and 2.64 at 10 ms for a 1 s memory. Distance 7 stops beating distance 3 near 100 ms. Nuclear memories mostly dephase; the standard code wastes this bias because its X memory fails first, while an XZZX code [24, 25], emulated by Hadamard conjugation of a checkerboard sublattice, pushes break-even to about 1000 ms (Fig. 4). Building the XZZX syndrome circuit natively (CZ and XCX gates on one sublattice, so hook errors are included) and giving the gates biased noise confirms this at the gate level: at bias $\eta = 100$ the standard code's threshold falls from 0.95% to 0.63% while the XZZX threshold rises to 1.59%, and for an NV cell at 1 ms readout the distance-7 XZZX memory is 12× better than the standard one (Fig. 5).

6 How big, how slow

Fitting $p_L(d) = A\Lambda^{-(d+1)/2}$ [26] and pricing each logical step at d rounds [27] gives, for an RSA-2048-scale job (6,000 logical qubits, 3×10^9 steps), distance 57, 78 million physical qubits on a 5.7 mm die, and 6.0 years at 1 ms readout (1.0 year at 0.1 ms), against 4 days for a superconducting machine at its measured $\Lambda = 2.14$ [28] (Fig. 6). If the gate noise is strongly biased (bias 100), the standard code needs distance 129 and 399 million qubits (13.6 years), while a native XZZX code needs distance 53, 67 million qubits, and 5.6 years: the code choice matters, although Section 8 shows XZZX is the better choice at every bias. The NV figure assumes a 0.3% link error that has not been demonstrated. Once readout is fast, the four dipolar gates dominate each round, so placement precision becomes the next lever.

7 Experiments and teaching

Ten experiments span US$100 to a pilot line (Fig. 7), following published teaching setups [29, 30, 31]. The cheapest ships with firmware (Raspberry Pi Pico driving an ADF4351 synthesizer) and a host script whose output feeds the same fitting code as a browser Data Lab. An eight-lesson course, 13 interactive labs, fourteen animated equations tied one-to-one to the experiments, and a process traveler turn the framework into teaching material, also packaged as an offline desktop app; all browser simulations are tested against the Python models.

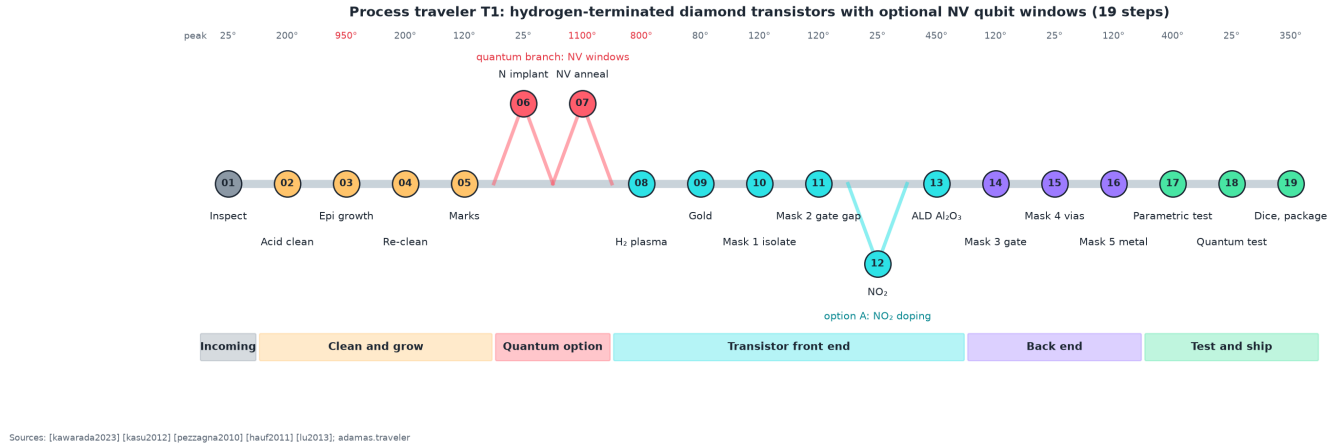


Figure 2: Process traveler T1: 19 steps from a bare plate to a tested chip, with a quantum branch (NV implant and anneal) and an NO_2 doping option. Heat is spent before the first metal.

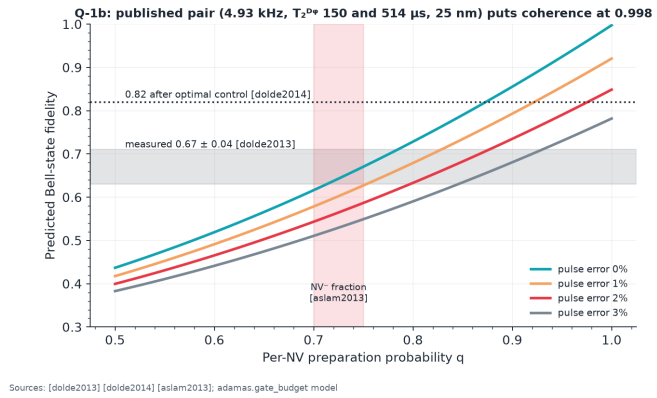


Figure 3: Echoed NV–NV gate budget evaluated with the published pair [16]: fidelity versus charge-state preparation probability. Coherence limits fidelity to 0.998.

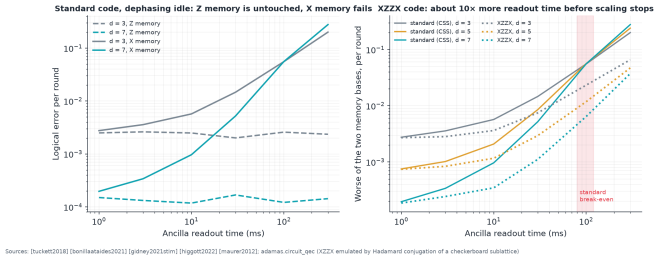


Figure 4: Circuit-level surface code with dephasing idle errors during readout. Left: standard code, Z and X memories. Right: worse basis for standard and XZZX codes.

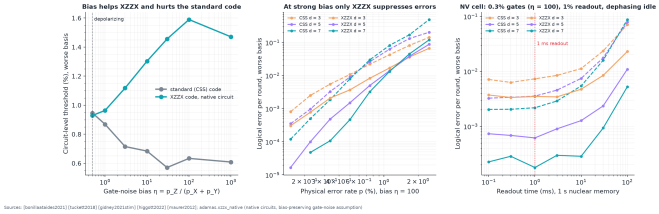


Figure 5: Native XZZX circuits with biased gate noise: thresholds versus bias, logical error at $\eta = 100$, and the NV readout scenario.

8 Validation and uncertainty

A validation matrix of 19 checks (18 pass, the rest explained discrepancies) compares each model with a published result, an independent tool, or a known answer; the phenomenological surface-code threshold, for example, reproduces the published 2.93% to within 6% [32]. Propagating the published spread of diamond's critical field and hole mobility [33, 34, 35] lowers the silicon-carbide-to-diamond loss ratio from the optimistic baseline to a median of 3.8× (P10 to P90: 2.5 to 5.5×). The runtime of the factoring-scale NV machine spans 1.2 to 15.5 years (median 3.6), dominated by readout time (Fig. 8). Variance-based (Sobol) sensitivity indices [36, 37, 38] make the ranking quantitative: diamond's hole mobility and critical field each explain about 0.42 and 0.42 of the power-ratio variance, and readout time 0.96 of the log runtime. Dividing by the cost of the experiment that pins each input down ranks a few-thousand-dollar Hall measurement of diamond hole mobility first; boron-nitride-gated hole channels near 1000 $\mathrm{cm}^2/\mathrm{V\,s}$ [39] show why the range is wide. Readout, the quantum side's dominant input, is improving through nuclear-assisted repetitive readout [40] and ionization-based initialization [41]. Finally, because the XZZX code needs fewer qubits than the standard code at every simulated gate bias (142 versus 165 million even without bias, with native circuits and dephasing idle), measuring the bias has zero expected value of information for the code choice [42]: it sizes the machine but does not change its code (Fig. 9).

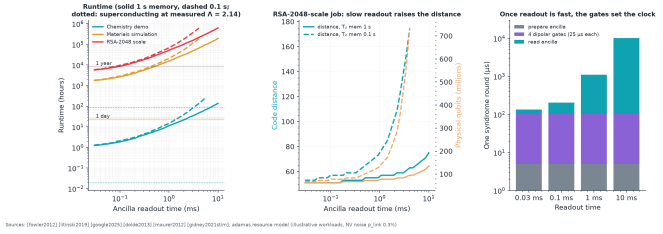


Figure 6: Resource estimate from the circuit-level fits: runtime, code distance, and the composition of one syndrome round.

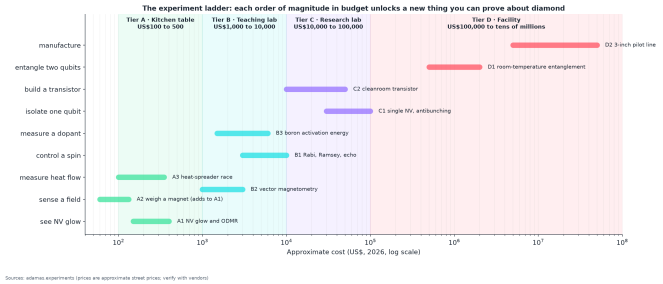


Figure 7: Ten experiments in four budgets, with parts, steps, safety, and simulated expected results; a firmware kit makes the US$150 to 400 ODMR experiment runnable.

9 Limitations

Transistor and switching parameters are literature-anchored placeholders until the PDK-0 monitor die is measured. The converter model omits gate charge and reverse recovery; the surface-code idle channel is phenomenological at the gate level (depolarizing or pure dephasing), and the native XZZX study assumes bias-preserving gate noise, which has not been measured for NV dipolar gates. Workloads in the resource estimate are illustrative scales.

Reproducibility

`pip install -e ".[dev]" && python -m pytest -q && make figures paper` regenerates every figure, the numbers in this text (`python -m adamas.paper`), and this PDF. Interactive version: normansrule.github.io/adamas-diamond-framework.

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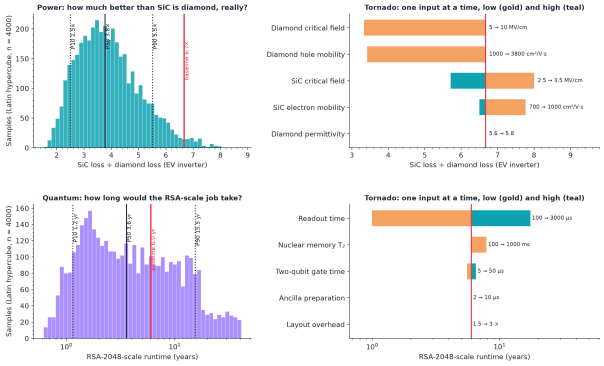


Figure 8: Latin-hypercube uncertainty propagation and tornado diagrams for the two headline results.

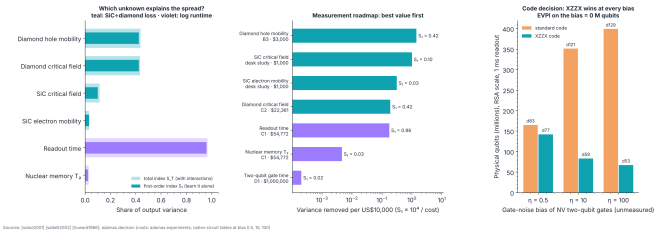


Figure 9: What to measure next: Sobol indices, variance removed per dollar, and the code decision under unknown gate bias.

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